1 Forward kinematics models

In this section, we present the forward kinematics model we obtained for our CuRo6R robot using Denavit–Hartenberg parameters and also using Product of Exponentials. We also show numerical agreement between the two models.

1.1 DH description

The six revolute joints are described using the *standard* DH convention. Frame $\{0\}$ coincides with `base_link`, and $q_i = 0$ denotes the URDF home configuration. The endpoint E is `link_6`. For each step i, the transform from frame $\{i-1\}$ to frame $\{i\}$ is

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i), \qquad \theta_i = q_i + \theta_i^0. \tag{1}$$

Joint i rotates about z_{i-1}. For $i < 6$, the new axis z_i lies on joint $i+1$; the terminal axis z_6 is chosen parallel to z_5. DH frame numbers therefore differ from physical joint numbers.

How to read one step. Choose x_i along a common normal to z_{i-1} and z_i. Its intersection with z_i defines O_i. If two distinct axes intersect, their common-normal length is zero and x_i is perpendicular to both axes through their intersection. For coincident axes, the origin and transverse x axis are free; here the last origin is the endpoint. Set $y_i = z_i \times x_i$ to complete a right-handed frame. The four parameters then have the following geometric meaning:

Parameter	Measured about / along	Construction from $\{i-1\}$ to $\{i\}$
θ_i	about z_{i-1}	Turn x_{i-1} towards x_i.
d_i	along z_{i-1}	Move from O_{i-1} to the foot of the common normal.
a_i	along x_i	Follow the common normal to O_i.
α_i	about x_i	Turn z_{i-1} towards z_i.

Translations are signed along the indicated positive axes; rotations follow the right-hand rule. Figure 3 shows the construction at $q = 0$.

Frame origins versus joint centres. A DH origin lies on the selected joint axis but need not coincide with a URDF joint origin. In particular, O_3 lies 1.5 m before the physical centre of joint 4. This distance enters d_4, giving $d_4 = 1.5 + 1 = 2.5$ m, while $a_3 = 0$. Similarly, $d_5 = 0$ even though the centres of joints 5 and 6 are 1.5 m apart: the two axes already intersect at $O_4 = O_5$.

Endpoint alignment. Frame $\{6\}$ and the URDF endpoint have the same origin but different axes: $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. Hence the forward kinematics uses the fixed alignment on the right:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

Table 1: Six DH steps. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric construction at home
1	1	q_1	$-\pi/3$	0.8	Rise $0.5+0.5$ from the base, then move 0.8 along x_1 to joint 2. Its axis gives a $-60°$ twist.
2	1.25	q_2	$+\pi/2$	2	Move 1.25 along joint 2's axis and 2 along x_2 to joint 3; twist by $+90°$.
3	0.75	$q_3+\pi/2$	$+\pi/2$	0	Extend joints 3 and 4's axes to their intersection O_3. Turn x_2 by $+90°$ to select x_3; no common-normal translation is needed.
4	2.5	q_4	$-\pi/2$	0	From O_3, follow joint 4's axis by $1.5+1$ to the wrist centre O_4; twist by $-90°$.
5	0	q_5	$+\pi/2$	0	Joints 5 and 6's axes meet at the wrist centre. Keep $O_5 = O_4$ and twist by $+90°$.
6	1.5	q_6	0	0	Follow joint 6's axis from the wrist centre to the endpoint. Choose $z_6 = z_5$ and no fixed twist.

1.2 Product of exponentials model

The product of exponentials (PoE) describes the robot through its joint axes at the home configuration. All vectors below are expressed in `base_link`, with $q = 0$ and endpoint `link_6` as in the DH model.

1. Describe each revolute joint by a screw axis. Let ω_i be the unit direction of joint i and let r_i be a point on its axis at home. With unit joint speed, a point p has velocity $\omega_i \times (p - r_i) = \omega_i \times p + v_i$. This gives the six-vector ξ_i and its matrix form:

$$v_i = -\omega_i \times r_i, \qquad \xi_i = \begin{pmatrix} \omega_i \\ v_i \end{pmatrix}, \qquad \widehat{\xi}_i = \begin{pmatrix} [\omega_i]_\times & v_i \\ 0^T & 0 \end{pmatrix}. \tag{3}$$

Here $[\omega]_\times$ represents the cross product:

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}, \qquad [\omega]_\times p = \omega \times p. \tag{4}$$

Any point on the same axis gives the same v_i: adding a multiple of ω_i to r_i does not change the cross product.

2. Find the six axes at home. The fixed tilt $\gamma = \pi/6$ maps the local y and z axes to $(0, \sqrt{3}/2, 1/2)^T$ and $(0, -1/2, \sqrt{3}/2)^T$. The points r_i follow by adding the URDF joint offsets in the base frame. For example, $r_3 = r_2 + (2, 0, 0)^T + (5/4)\omega_2$. Applying (3) then gives Table 2.

At home, axes 4 and 6 lie on the same line and point in the same direction. Their chosen origins differ only along that line, so $\xi_4 = \xi_6$. This equality describes the home configuration; the axes need not remain aligned when joint 5 moves.

3. Obtain the motion of one joint. For a unit axis ω_i, Rodrigues' formula gives the

Table 2: Home screw data. Each vector is written as a row; r_i uses the physical joint origin. Lengths are in metres.

i	ω_i^T	r_i^T	v_i^T
1	$(0, 0, 1)$	$(0, 0, \frac{1}{2})$	$(0, 0, 0)$
2	$(0, \frac{\sqrt{3}}{2}, \frac{1}{2})$	$(\frac{4}{5}, 0, 1)$	$(-\frac{\sqrt{3}}{2}, -\frac{2}{5}, \frac{2\sqrt{3}}{5})$
3	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{14}{5}, \frac{5\sqrt{3}}{8}, \frac{13}{8})$	$(\frac{7}{4}, -\frac{7\sqrt{3}}{5}, -\frac{7}{5})$
4	$(1, 0, 0)$	$(\frac{43}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$
5	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{53}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(\frac{7}{4}, -\frac{53\sqrt{3}}{20}, -\frac{53}{20})$
6	$(1, 0, 0)$	$(\frac{34}{5}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$

rotation $R_i(q_i)$. The corresponding rigid motion is the matrix exponential

$$\begin{aligned} R_i(q_i) &= I_3 + \sin q_i [\omega_i]_\times + (1 - \cos q_i)[\omega_i]_\times^2, \\ e^{\widehat{\xi}_i q_i} &= \begin{pmatrix} R_i(q_i) & (I_3 - R_i(q_i))r_i \\ 0^T & 1 \end{pmatrix}. \end{aligned} \tag{5}$$

The translation term follows by writing the rotated point as $r_i + R_i(q_i)(p - r_i)$. It keeps every point on the joint axis fixed.

4. Set the home pose and combine the joint motions. At $q = 0$, the endpoint has orientation $R_x(\gamma)$ and position r_6. Its home pose is therefore

$$M = {}^0T_E(0) = \begin{pmatrix} 1 & 0 & 0 & \frac{34}{5} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{5\sqrt{3}-3}{8} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{13+3\sqrt{3}}{8} \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6}$$

The complete forward kinematics is

$$\boxed{{}^0T_E(q) = e^{\widehat{\xi}_1 q_1} e^{\widehat{\xi}_2 q_2} \cdots e^{\widehat{\xi}_6 q_6} M.} \tag{7}$$

The factors are written in joint order, from 1 to 6. The screw axes are fixed at their home values; the product accounts for how earlier joints move the later axes. At $q = 0$, all exponentials are identity matrices and the expression returns M.

1.3 Verification

Using the data above, the PoE, DH and direct URDF models were compared at 514 configurations: home, each joint alone at $\pm\pi/2$, one mixed pose, and 500 random configurations in $[-\pi, \pi]^6$. Across all model pairs, the pose-matrix Frobenius residuals were below 10^{-14}. The MATLAB URDF transcription uses the exact tilt $\pi/6$; a separate read-only check compares its joint placements, axes and chain with the supplied URDF, allowing 10^{-10} for decimal truncation.

2 Jacobian and singularities

In this section we construct the geometric Jacobian of CuRo6R from the PoE model. The MATLAB PoE code keeps the intermediate transforms. These give each joint's current unit direction w_i and a point p_i on its axis, both in `base_link`. For a revolute joint, the Jacobian column is

$$ {}^0J_k(:,i) = \begin{pmatrix} w_i \times (p_k - p_i) \\ w_i \end{pmatrix}, \qquad i = 1, \ldots, 6. \tag{8} $$

The upper rows give linear velocity and the lower rows give angular velocity. We use $k = 6$ for the endpoint. Choosing $k = 5$ instead refers the same tool motion to the wrist centre.

Arm and wrist blocks. Here ${}^3J_5$ denotes this full Jacobian at the wrist centre, expressed along the physical link-3 axes (`link_3_d3`). Both velocity blocks are rotated by $({}^0R_3)^T$; the velocities remain relative to the base. All three wrist axes pass through p_5, so their linear columns vanish. With $c_i = \cos q_i$ and $s_i = \sin q_i$, the result has the form

$$ {}^3J_5 = \begin{pmatrix} J_{v,\mathrm{arm}} & 0 \\ J_{\omega,\mathrm{arm}} & J_{\omega,\mathrm{wrist}} \end{pmatrix}, \qquad J_{\omega,\mathrm{wrist}} = \begin{pmatrix} 1 & 0 & c_5 \\ 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \end{pmatrix}. \tag{9} $$

The subscripts v and ω denote linear and angular velocity. The 3×3 blocks $J_{v,\mathrm{arm}}$ and $J_{\omega,\mathrm{arm}}$ map the rates of joints 1–3 to these two velocities; $J_{\omega,\mathrm{wrist}}$ maps the rates of joints 4–6 to angular velocity. Their linear contribution at the wrist centre is zero. This block triangular form makes the determinant easy to compute.

Symbolic and numerical routes. The same MATLAB function accepts numerical or symbolic inputs. Numerical inputs evaluate the Jacobian at a chosen configuration. Symbolic inputs give an exact expression, from which the determinant is obtained. We use $J = {}^3J_5$, whose zero block makes the determinant easy to evaluate:

$$ \det J(q) = -\sin q_5 \, \det J_{v,\mathrm{arm}}(q_2, q_3). \tag{10} $$

Using the DH lengths, define $U = a_2 + d_4 c_3$, $X = U c_2 + d_3 s_2$, and $Z = -U s_2 + d_3 c_2$. The explicit arm determinant is

$$ \det J_{v,\mathrm{arm}} = -\frac{d_4}{2} \left[a_1 Z c_3 - \sqrt{3}\, a_1 U s_3 - \sqrt{3}\, X (a_2 s_3 - d_2 c_3) \right]. \tag{11} $$

Changing the expression frame or velocity reference point preserves $\det J$, provided all six columns describe the same full tool motion. The wrist factor vanishes at $q_5 = 0$ or π modulo 2π. Axes 4 and 6 then lie on the same line, so one independent rotation is lost. The arm factor depends jointly on q_2 and q_3. Thus the wrist and arm conditions separate, while the arm condition must be studied in the (q_2, q_3) plane, as shown in Figure 1.

A full configuration is regular exactly when both factors are nonzero. Numerically, we monitor the smallest singular value of J. It becomes zero at a singularity, and small values indicate that the Jacobian is close to losing rank.

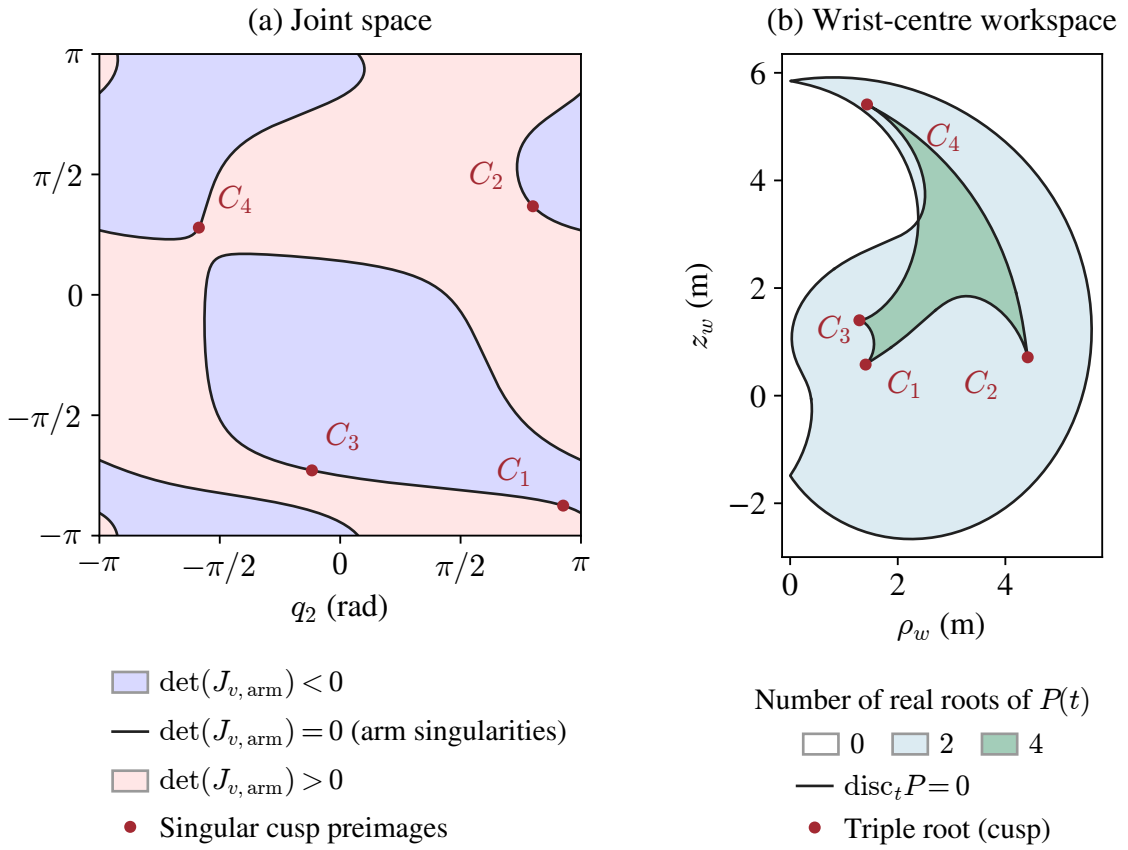


Figure 1: Arm singularities in (a) joint space and (b) the wrist-centre workspace, with $\rho_w = \sqrt{x_w^2 + y_w^2}$. In (a), colours give the sign of $\det J_{v,\text{arm}}$ and black curves show $\det J_{v,\text{arm}} = 0$. In (b), these curves map to real multiple roots of the IK quartic $P(t)$ in (28), derived in Subsection 3.4, where $\text{disc}_t\, P = 0$. Colours give the number of distinct real roots away from the curves, including the half-angle point at infinity when needed; for $\rho_w > 0$, this is the number of arm solutions before joint limits. Matching labels mark four cusps (triple roots) and their joint-space preimages. They show that the robot's positioning chain is cuspidal: it can change between IK solutions without crossing an arm singularity. Opposite joint-space edges match modulo 2π; wrist singularities remain a separate condition.

The physical link-3 frame has home orientation $R_x(\pi/6)$ and follows the first three joint motions. In its axes, the wrist columns are e_x, $R_x(q_4)e_z$, and $R_x(q_4)R_z(q_5)e_x$, giving the block $J_{\omega,\text{wrist}}$ in (9).

How the intermediate PoE transforms give the axes. Write the product before joint i as $F_{i-1} = e^{\widehat{\xi}_1 q_1} \cdots e^{\widehat{\xi}_{i-1} q_{i-1}} = (R_{i-1}, t_{i-1})$. Then $w_i = R_{i-1}\omega_i$ and $p_i = R_{i-1}r_i + t_{i-1}$. The current screw's linear component is $-w_i \times p_i$. Evaluating its velocity field at p_k adds $w_i \times p_k$, giving the upper block of (8).

Why the determinant separates. Since $\det J_{\omega,\text{wrist}} = -\sin q_5$, the block triangular form in (9) gives the factorization (10). Expressing both velocity blocks in another rotated frame multiplies the Jacobian by $\text{diag}(R^T, R^T)$, whose determinant is one. If $d = p_\ell - p_j$, changing the velocity reference gives

$$ {}^0J_\ell = \begin{pmatrix} I_3 & -[d]_\times \\ 0 & I_3 \end{pmatrix} {}^0J_j. \tag{12} $$

This matrix also has determinant one. Thus $\det({}^{i}J_j) = \det({}^{3}J_5)$ for every expression frame i and velocity reference point p_j, when all six columns describe the same full tool motion. This is why ${}^{3}J_5$ suffices to study the full robot's singularities. Changing the reference point preserves rank, but can change the singular values. Numerical singular-value margins should therefore use a fixed reference point.

From joint space to the wrist-centre workspace. Set $V = d_2 + d_4 s_3$, $c_\gamma = \cos\gamma$ and $s_\gamma = \sin\gamma$. At $q_1 = 0$, the wrist centre is

$$p_w = \begin{pmatrix} a_1 + X \\ c_\gamma V - s_\gamma Z \\ d_1 + s_\gamma V + c_\gamma Z \end{pmatrix}, \qquad \rho_w = \sqrt{(a_1 + X)^2 + (c_\gamma V - s_\gamma Z)^2}. \tag{13}$$

Changing q_1 rotates this point about the base z-axis, leaving (ρ_w, z_w) unchanged. The right panel of Figure 1 maps each point of $\det J_{v,\mathrm{arm}} = 0$ through these equations. It shows the wrist centre, whose position depends on joints 1–3. The tool endpoint also depends on the wrist orientation through $p_E = p_w + d_6 R_E e_x$.

Checking the cusps. The four marked points were checked using the arm IK polynomial $P(t)$ in (28). For each wrist point, form the target invariants $h = z_w - d_1$ and $\mathcal{R} = x_w^2 + y_w^2 + h^2$. At $t = \tan(q_3/2)$, each marked point satisfies

$$P(t) = P'(t) = P''(t) = 0, \qquad P'''(t) \neq 0. \tag{14}$$

Thus three arm IK solutions coincide at each cusp. After normalising the polynomial coefficients, the largest residual in the three zero conditions was 4.0×10^{-15}. The cusp criterion establishes that the positioning chain is cuspidal (see Salunkhe et al., 2022). This means that some arm IK solutions can be connected without crossing an arm singularity. A motion of the full robot must also keep $\sin q_5 \neq 0$ and respect the joint limits.

3 Analytical inverse kinematics

3.1 Frames, geometry and the wrist target

We seek the joint vectors $\mathbf{q} = (q_1, \ldots, q_6)^{\mathsf{T}}$ giving a prescribed pose of `link_6`. Frame 0 is `base_link`; frames $1, \ldots, 6$ are the URDF link frames. The q_i are URDF joint displacements from home.

Let C be the wrist centre, the origin of frame 5. The geometry is specified by

$$\begin{aligned} {}^{0}\mathbf{t}_2\big|_{q_1=0} &= \begin{pmatrix} a_1 \\ 0 \\ d_1 \end{pmatrix}, & {}^{2}\mathbf{t}_3 &= \begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}, \\ {}^{3}\mathbf{p}_C &= \begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}, & Q &:= {}^{1}R_2\big|_{q_2=0} = R_x(\alpha). \end{aligned} \tag{15}$$

Here d_1 combines the two vertical base offsets, and d_4 combines the x offsets from joint 3 to joint 4 and from joint 4 to joint 5. For the supplied robot, $(a_1, a_2, d_1, d_2, d_3, d_4, d_6) = (0.8, 2, 1, 1.25, 0.75, 2.5, 1.5)\,\mathrm{m}$ and $\alpha = \pi/6$. Write $c_i = \cos q_i$, $s_i = \sin q_i$, $c_\alpha = \cos\alpha$

and $s_\alpha = \sin\alpha$. The matrices R_x, R_y, R_z are right-handed rotations; $\mathbf{e}_x = (1,0,0)^\mathsf{T}$ and $\mathbf{e}_z = (0,0,1)^\mathsf{T}$ are coordinate unit vectors. The two-argument function atan2(s,c) denotes the four-quadrant inverse tangent, with the sine argument first.

All three wrist axes pass through C, so its position depends only on q_1, q_2, q_3. The origin of frame 6 lies a distance d_6 from C along its own x axis. The subscript d denotes a desired quantity, so the target pose gives the desired wrist position

$$\boxed{{}^0\mathbf{p}_{C,d} = {}^0\mathbf{t}_{6,d} - d_6\,{}^0R_{6,d}\mathbf{e}_x = \begin{pmatrix} x_d & y_d & z_d \end{pmatrix}^\mathsf{T}}. \tag{16}$$

3.2 Writing the wrist-centre position

Combining the relative translations and rotations gives us

$${}^0\mathbf{p}_C(\mathbf{q}) = \underbrace{d_1\mathbf{e}_z}_{{}^0\mathbf{p}_O} + \underbrace{R_z(q_1)}_{{}^0R_1} \underbrace{\Bigg[\underbrace{a_1\mathbf{e}_x}_{{}^1\mathbf{t}_2 - {}^1\mathbf{p}_O} + \underbrace{QR_y(q_2)}_{{}^1R_2} \underbrace{\Bigg(\underbrace{\begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}}_{{}^2\mathbf{t}_3} + \underbrace{R_z(q_3)}_{{}^2R_3} \underbrace{\begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}}_{{}^3\mathbf{p}_C} \Bigg)}_{{}^2\mathbf{p}_C} \Bigg]}_{{}^1\mathbf{p}_C - {}^1\mathbf{p}_O}. \tag{17}$$

This is also the position part of the product of exponentials. We evaluate it from the inside out. First, joint 3 gives us the shoulder-to-wrist coordinates in frame 2: ${}^2\mathbf{p}_C = (F_1, B, F_2)^\mathsf{T}$, where

$$F_1 := a_2 + d_4 c_3, \qquad B := d_2 + d_4 s_3, \qquad F_2 := d_3. \tag{18}$$

Next we apply joint 2, then the fixed tilt and the shoulder offset:

$$\begin{aligned} \underbrace{\begin{pmatrix} X \\ B \\ D \end{pmatrix}}_{\text{before the tilt}} &= R_y(q_2) \begin{pmatrix} F_1 \\ B \\ F_2 \end{pmatrix} = \begin{pmatrix} F_1 c_2 + F_2 s_2 \\ B \\ F_2 c_2 - F_1 s_2 \end{pmatrix}, \\ \underbrace{\begin{pmatrix} A \\ Y \\ H \end{pmatrix}}_{\text{before joint 1}} &= a_1\mathbf{e}_x + Q \begin{pmatrix} X \\ B \\ D \end{pmatrix} = \begin{pmatrix} a_1 + X \\ c_\alpha B - s_\alpha D \\ s_\alpha B + c_\alpha D \end{pmatrix}. \end{aligned} \tag{19}$$

A, Y, H are the wrist's two horizontal components and its height relative to the point on axis 1 at height d_1, in axes that rotate with joint 1. Applying the final base rotation gives us

$$x_d = Ac_1 - Y s_1, \qquad y_d = As_1 + Y c_1, \qquad z_d = d_1 + H. \tag{20}$$

3.3 Eliminating the first two joint angles

Let O be the point on axis 1 at shoulder height, ${}^0\mathbf{p}_O = (0,0,d_1)^\mathsf{T}$. We now form three *known target quantities*:

$$h := z_d - d_1, \qquad \rho^2 := x_d^2 + y_d^2, \qquad \mathcal{R} := \rho^2 + h^2 = \|{}^0\mathbf{p}_{C,d} - {}^0\mathbf{p}_O\|^2. \tag{21}$$

h is the wrist height above O, ρ is wrist distance to axis 1, and $\mathcal{R}$ is the wrist squared distance to O. None changes when joint 1 rotates. For a solution, (20) gives $H = h$ and $\rho^2 = A^2 + Y^2$.

Rotation preserves norm. We apply that to the first relation in (19):

$$\underbrace{X^2 + B^2 + D^2}_{\|R_y(q_2)(F_1,B,F_2)^\mathsf{T}\|^2} = \underbrace{F_1^2 + B^2 + F_2^2}_{\|(F_1,B,F_2)^\mathsf{T}\|^2} \quad \Longrightarrow \quad X^2 + D^2 = F_1^2 + F_2^2. \tag{22}$$

The unchanged y component B cancels. Likewise, $(Y, H)^\mathsf{T} = \left(\begin{smallmatrix} c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha \end{smallmatrix}\right)(B, D)^\mathsf{T}$ gives $Y^2 + h^2 = B^2 + D^2$. Hence

$$\begin{aligned} &\mathcal{R} = (a_1 + X)^2 + B^2 + D^2 = 2a_1 X + F_3, \qquad h = s_\alpha B + c_\alpha D, \\ &F_3 := a_1^2 + F_1^2 + F_2^2 + B^2 = L + 2a_2 d_4 c_3 + 2d_2 d_4 s_3, \\ &L := a_1^2 + a_2^2 + d_2^2 + d_3^2 + d_4^2. \end{aligned} \tag{23}$$

The expansion of F_3 uses $c_3^2 + s_3^2 = 1$: F_3 depends only on q_3, while L is constant. With $\mathcal{R}, h$ known and a_1, c_α nonzero, a trial q_3 fixes

$$X = \frac{\mathcal{R} - F_3}{2a_1}, \qquad D = \frac{h - s_\alpha B}{c_\alpha}. \tag{24}$$

These required X, D can come from a real rotation q_2 only if their norm is correct. Substituting into (22) and multiplying by $4a_1^2$ gives us the *closure condition*:

$$\boxed{G(q_3) := (\mathcal{R} - F_3)^2 + \frac{4a_1^2}{c_\alpha^2}(h - s_\alpha B)^2 - 4a_1^2(F_1^2 + d_3^2) = 0.} \tag{25}$$

Only q_3 remains unknown. For $\alpha = 0$, $D = h$ and this becomes the lecture's untilted closure, with terminal arm length d_4.

3.4 Converting the closure into a quartic

We use the half-angle substitution

$$t = \tan(q_3/2), \qquad \Delta = 1 + t^2, \qquad c_3 = \frac{1 - t^2}{\Delta}, \qquad s_3 = \frac{2t}{\Delta}. \tag{26}$$

The hat below denotes a numerator after substitution. Multiplying by the common denominator Δ gives

$$\begin{aligned} \widehat{F}_1(t) &:= \Delta F_1 = a_2\Delta + d_4(1 - t^2), \\ \widehat{B}(t) &:= \Delta B = d_2\Delta + 2d_4 t, \\ \widehat{F}_3(t) &:= \Delta F_3 = L\Delta + 2a_2 d_4(1 - t^2) + 4d_2 d_4 t. \end{aligned} \tag{27}$$

Each numerator is quadratic in t. Multiplying (25) by $c_\alpha^2\Delta^2$ clears the denominators and yields a polynomial of degree at most four, because it contains squares of these quadratics:

$$\boxed{\begin{aligned} P(t) = c_\alpha^2(\mathcal{R}\Delta - \widehat{F}_3)^2 + 4a_1^2(h\Delta - s_\alpha\widehat{B})^2 \\ - 4a_1^2 c_\alpha^2(\widehat{F}_1^2 + d_3^2\Delta^2) = 0. \end{aligned}} \tag{28}$$

Every real root of the polynomial gives a $q_3 = 2\arctan t$. Since $\Delta > 0$ for real t, clearing denominators adds no real roots. We must test $q_3 = \pi$ separately in (25), because its half-angle tangent is undefined. This missing point is included in the maximum of four distinct q_3 candidates.

3.5 Recovering the remaining arm angles

For each candidate q_3, we evaluate (18) for F_1, F_2, B and use (24) for X, D. Combining their defining equations cancels s_2 in $F_1X + F_2D$ and c_2 in $F_2X - F_1D$, giving us

$$c_2 = \frac{F_1X + F_2D}{F_1^2 + F_2^2}, \qquad s_2 = \frac{F_2X - F_1D}{F_1^2 + F_2^2}, \qquad q_2 = \operatorname{atan2}(s_2, c_2). \tag{29}$$

The denominator is positive because $F_2 = d_3 \neq 0$, and the closure ensures $c_2^2 + s_2^2 = 1$. Next we set $A = a_1 + X$, $Y = c_\alpha B - s_\alpha D$ and recover the base angle from the horizontal equations:

$$q_1 = \operatorname{atan2}(\underbrace{Ay_d - Yx_d}_{\rho^2 s_1}, \underbrace{Ax_d + Yy_d}_{\rho^2 c_1}), \qquad \rho > 0. \tag{30}$$

The common positive factor ρ^2 does not change the angle. If $\rho = 0$, the wrist centre lies on axis 1 and q_1 is free, in which case we choose a value in $[-\pi, \pi]$ and solve the wrist for that choice.

3.6 Recovering the wrist angles

For each arm solution, we undo its orientation using the transpose, then solve the remaining local x–z–x wrist rotations:

$$\begin{aligned} {}^0R_3 &= R_z(q_1) Q R_y(q_2) R_z(q_3), \\ {}^3R_{6,d} &= ({}^0R_3)^\mathsf{T}\, {}^0R_{6,d} = R_x(q_4) R_z(q_5) R_x(q_6) =: (r_{ij}). \end{aligned} \tag{31}$$

Here r_{ij} is entry (i, j) of ${}^3R_{6,d}$. Multiplying the wrist rotations gives us five useful entries:

$$r_{11} = c_5, \quad r_{21} = c_4 s_5, \quad r_{31} = s_4 s_5, \quad r_{12} = -s_5 c_6, \quad r_{13} = s_5 s_6. \tag{32}$$

Let $\eta = \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\eta > 0$, we choose each sign $\varepsilon \in \{-1, +1\}$, so that $s_5 = \varepsilon\eta$. Then

$$\boxed{\begin{aligned} &q_5 = \operatorname{atan2}(\varepsilon\eta, r_{11}), \\ &q_4 = \operatorname{atan2}(\varepsilon r_{31}, \varepsilon r_{21}), \qquad q_6 = \operatorname{atan2}(\varepsilon r_{13}, -\varepsilon r_{12}). \end{aligned}} \tag{33}$$

We must use the same sign in all three formulas. It selects one of two wrist configurations with the same orientation. If $\eta = 0$, joints 4 and 6 are coaxial, so only a sum or difference is fixed:

$$\begin{aligned} r_{11} = 1: &\quad q_5 = 0, \quad q_4 + q_6 = \operatorname{atan2}(r_{32}, r_{22}), \\ r_{11} = -1: &\quad q_5 = \pi, \quad q_6 - q_4 = \operatorname{atan2}(r_{32}, -r_{22}). \end{aligned} \tag{34}$$

We may set $q_4 = 0$ and recover q_6 from the appropriate relation.

3.7 Assembling and checking the solutions

For every real q_3 candidate, we recover the arm and both wrist branches. We represent angles in $[-\pi, \pi]$, enforce joint limits, remove duplicates modulo 2π, and check position and orientation by forward kinematics. We use numerical tolerances for roots, singularities and pose errors. Away from continuous singular families, at most four arm solutions each have two wrist branches, so there are **at most eight full-pose solutions**.

4 Connectivity analysis

A nonsingular change of solution returns the tool to its initial pose while leaving the robot in a different joint configuration and keeping the full pose regular throughout the motion. In this section, we show how a nonsingular change of solution can be achieved.

4.1 A straight arm path with fixed tool orientation

We first test a simple construction: connect two arm solutions by a straight segment in (q_1, q_2, q_3), then adjust the wrist to keep the tool orientation fixed. The tool position is free to move between the endpoints.

Selecting a candidate pair. For the initial configuration $\mathbf{q}^A$ below, the analytical solver returns eight solutions of $T_d = \mathrm{FK}(\mathbf{q}^A)$. Equation (10) gives two necessary conditions for a regular connection: $\det J_{v,\mathrm{arm}}$ and the wrist factor $\sin q_5$ must each retain their sign. Equal signs alone do not establish connectivity; the path between the endpoints must still be checked. After removing the initial solution and screening the signs and endpoint margins, one candidate remains, $\mathbf{q}^B$. In radians,

$$\begin{aligned}(\mathbf{q}^A)^\mathsf{T} &= (\ \ 0.1024,\ -1.1124,\ \ \ 1.8547,\ -1.4399,\ -0.3835,\ -2.6486),\\ (\mathbf{q}^B)^\mathsf{T} &\simeq (-2.2287,\ -2.1482,\ -0.2788,\ \ \ 0.6181,\ -0.3524,\ -1.7072).\end{aligned} \tag{35}$$

These are solutions 8^- and 4^- in Figure 2; the superscript denotes $\mathrm{sign}(\sin q_5)$.

From the arm segment to the full motion. Writing $\mathbf{q}_a = (q_1, q_2, q_3)^\mathsf{T}$, we prescribe

$$\mathbf{q}_a(s) = (1-s)\mathbf{q}_a^A + s\mathbf{q}_a^B, \qquad 0 \le s \le 1, \tag{36}$$

where s is a path advancement parameter. At each sample, we compute $({}^0R_3(\mathbf{q}_a(s)))^\mathsf{T} R_d$ and recover q_4, q_5, q_6 using (33), always choosing the initial wrist sign. The angles are unwrapped towards the previous sample before testing $-\pi \le q_i \le \pi$; wrapping an angle back into this interval would hide a physical joint-limit crossing. With $\mathbf{p}_C$ the wrist centre, the resulting tool path is $\mathbf{p}_6(s) = \mathbf{p}_C(\mathbf{q}_a(s)) + d_6 R_d \mathbf{e}_x$. Both arm endpoints reach the same wrist centre, so $\mathbf{p}_6(1) = \mathbf{p}_6(0)$, although $\mathbf{q}(1) \neq \mathbf{q}(0)$. Thus a straight arm segment produces a closed, curved workspace path.

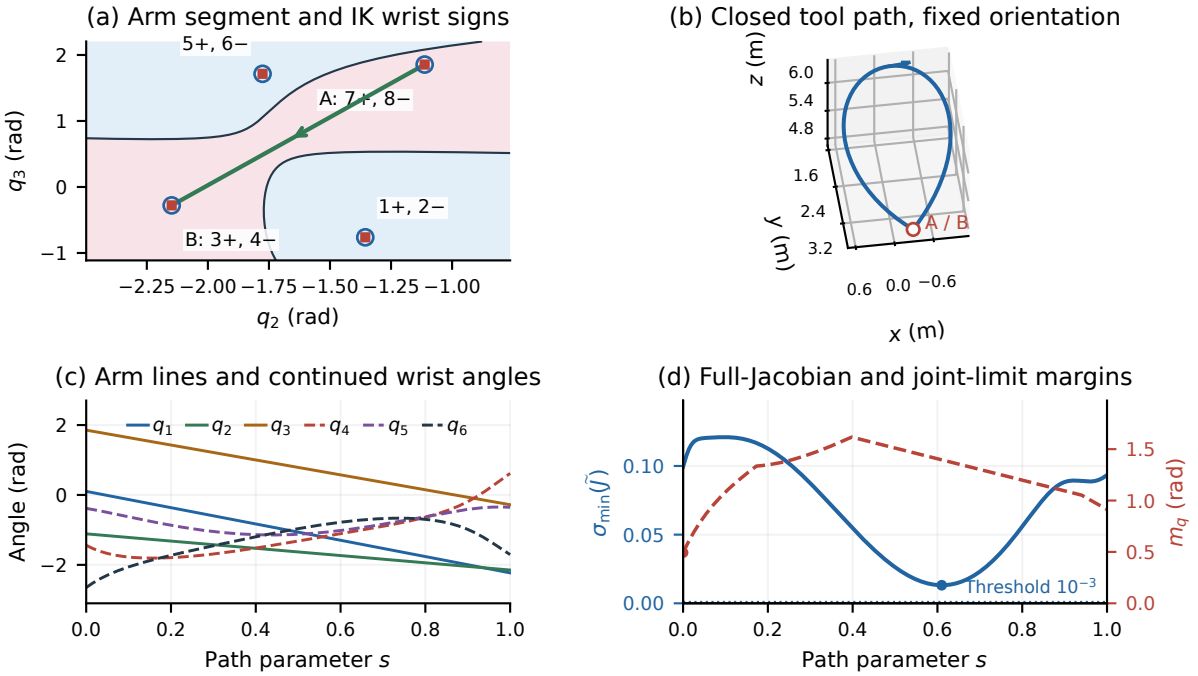


Figure 2: A fixed-orientation change from $A = 8^-$ to $B = 4^-$. In (a), black curves are $\det J_{v,\mathrm{arm}} = 0$; blue and pink denote negative and positive values, respectively. Each arm point carries both wrist signs, shown by a circle (+) and square (−). The green segment stays in one regular arm region. Panels (b)–(d) show the tool loop, all six joint paths, and the sampled margins.

Validation and interpretation. We evaluate 10 000 equally spaced samples, monitoring the arm sign, $|\sin q_5| > 10^{-6}$ and the full tool Jacobian. To compare its linear and angular rows, we use a characteristic length $L = 2\,\mathrm{m}$:

$$\tilde{J} = \mathrm{diag}(L^{-1}I_3, I_3)\,{}^0J_6, \qquad m_q(s) = \min_i\{q_i(s) + \pi,\ \pi - q_i(s)\}. \tag{37}$$

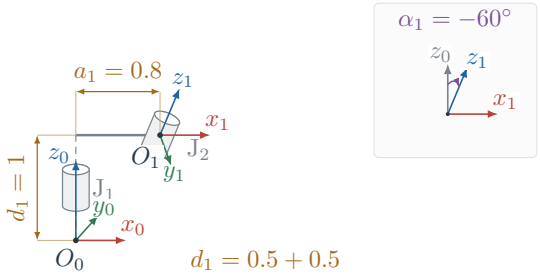
The minimum $\sigma_{\min}(\tilde{J})$ is 0.0132, above the 10^{-3} threshold; the minimum joint clearance is 0.493 rad and $\min|\sin q_5| = 0.337$. Position closure is 2.22×10^{-15} m and the maximum orientation error is 2.11×10^{-8} rad. Independent DH evaluation agrees with the PoE positions within 5.50×10^{-15} m. This numerically demonstrates a solution change for the selected pair. The checks concern sampled configurations, not every intervening point. A rejected segment would identify an arm, wrist or joint-limit obstruction along that attempt; it would not rule out a curved connection.

A DH frame constructions

Figure 3 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

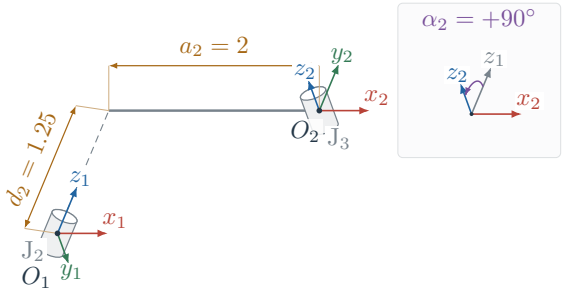
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

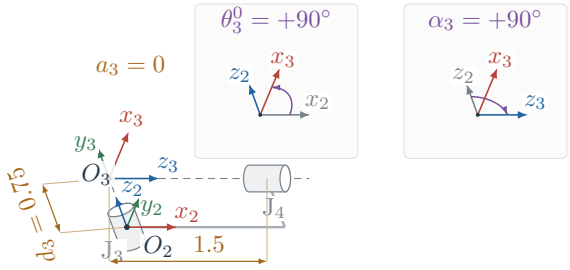
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

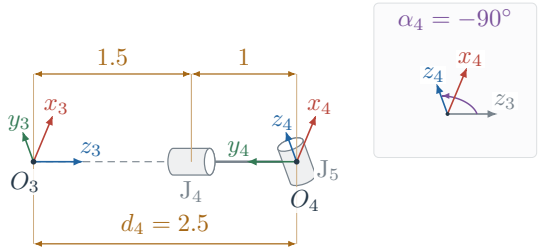
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

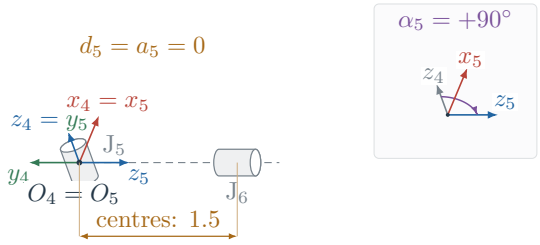
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

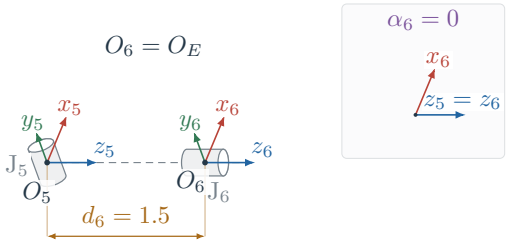
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 3: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.